

EXAMPLE 3.1

Outcomes

- DSTWU distillation modelling
- Reflux Ratio versus Number of Stages
- Plot graphs

Problem Description

A stream containing 68.5 wt.% ethylene and 31.5 wt.% ethane with a total flowrate of 7.3 million kg/day must be separated. Report a reasonable design parameters estimation using DSTWU model block required to achieve a separation of 99.9% recovery of ethylene and 99.0% recovery of ethane. Use Peng-Rob thermodynamic model.

1. Start a new simulation using the Blank Simulation template in Aspen Plus.
2. The Components | Specifications | Selection sheet is displayed. Enter ETHYLENE in the Component ID column. Note that Component name and Alias are automatically filled for this component. Enter ETHANE as Component ID for the second component.

Component ID	Type	Component name	Alias	CAS number
ETHYLENE	Conventional	ETHYLENE	C2H4	74-85-1
ETHANE	Conventional	ETHANE	C2H6	74-84-0

3. Define property methods. Click on Methods in the navigation pane. Select Peng-Rob as the Base Method. The Peng-Robinson equation of state is typically used to model systems containing hydrocarbons at high pressures.

Method name: PENG-ROB

Method filter: COMMON

Base method: PENG-ROB

Henry components: NONE

Petroleum calculation options:

- Free-water method: STEAM-TA
- Water solubility: 3

Electrolyte calculation options:

- Chemistry ID:
- ☒ Use true components

Modify:

- EOS: EPRSTD
- Data set: 1
- Liquid gamma: NONE
- Data set: 1
- Liquid molar enthalpy: HLMX106
- Liquid molar volume: VLMX20
- ☐ Heat of mixing
- ☐ Poynting correction
- ☐ Use liquid reference state enthalpy

- Click NEXT to retrieve the binary interaction parameters from Aspen database.

Binary Interaction - PRKBV-1 (T-DEPENDENT) x +

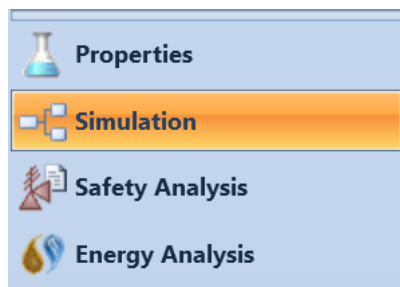
Input Databanks Comments

Parameter PRKBV Help Data set 1 Swap Search BIP Completeness

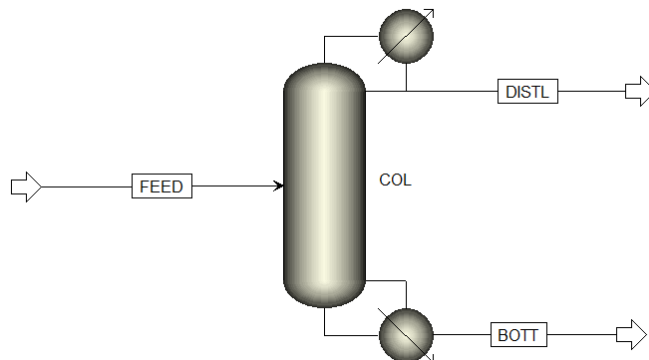
Temperature-dependent binary parameters

Component i	Component j	Source	Temp. Units	KAIJ	KBIJ	KCIJ	TLOWER	TUPPER
ETHYLENE	ETHANE	APV121 EOS-LIT	C	0.0089	0	0	-273.15	726.85
*								

- Go to the simulation environment by clicking on the Simulation button below the navigation pane.



- To create the process flowsheet, add the DSTWU model located under the Columns tab in the Model Palette, and name it COL (right-click on the stream and click rename or select the stream and press Ctrl+M). Connect the inlet and outlet ports with Material streams and name them accordingly.



- Double-click the FEED stream to enter the input stream characteristics. Select Vapor Fraction and Pressure for Flash Type. Enter 25 bar for Pressure and 1 for Vapor fraction. Change the Total flow basis to Mass and the units to kg/day. Enter 7,300,000 into the Total flow rate field. Change the Composition basis to Mass-Frac and enter 0.315 for ethane and 0.685 for ethylene.

☒ Mixed
 ☐ Cl Solid
 ☐ NC Solid
 ☐ Flash Options
 ☐ EO Options
 ☐ Costing
 ☐ Comments

Specifications

Flash Type: **Pressure** **Vapor Fraction**

State variables

Temperature: C

Pressure: **25** **bar**

Vapor fraction: **1**

Total flow basis: **Mass**

Total flow rate: **7.3e+06** **kg/day**

Solvent:

Reference Temperature

Volume flow reference temperature: C

Component concentration reference temperature: C

Composition: **Mass-Frac**

Component	Value
ETHYLENE	0.685
ETHANE	0.315

Total: **1**

8. Double-click on the COLUMN block to enter the specifications. Enter an initial guess for the Number of stages. The Light key is ethylene, and the Heavy key is ethane. Enter the desired recoveries for these components in the distillate stream, 99.9% and 1%, respectively. The condenser and reboiler pressure should both be 21 bar.

Main Flowsheet x COL (DSTWU) x +

☒ Specifications
 ☐ Calculation Options
 ☐ Convergence
 ☐ Comments

Column specifications

☒ Number of stages: **30**
☐ Reflux ratio:

Pressure

Condenser: **21** **bar**

Reboiler: **21** **bar**

Key component recoveries

Light key

Comp: **ETHYLENE**

Recov: **0.999**

Heavy key

Comp: **ETHANE**

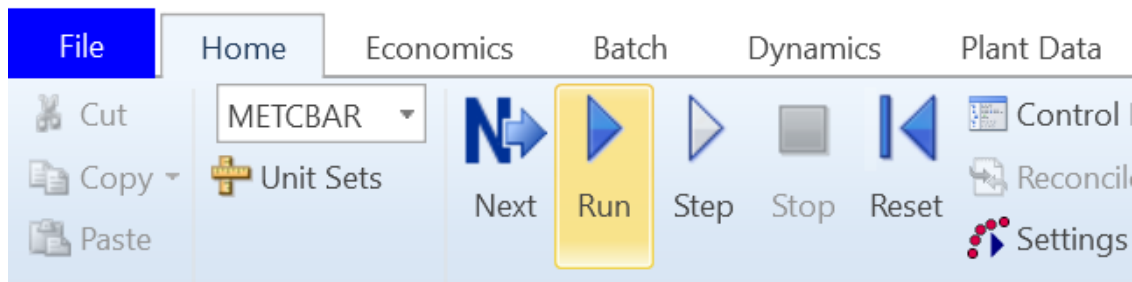
Recov: **0.01**

Condenser specifications

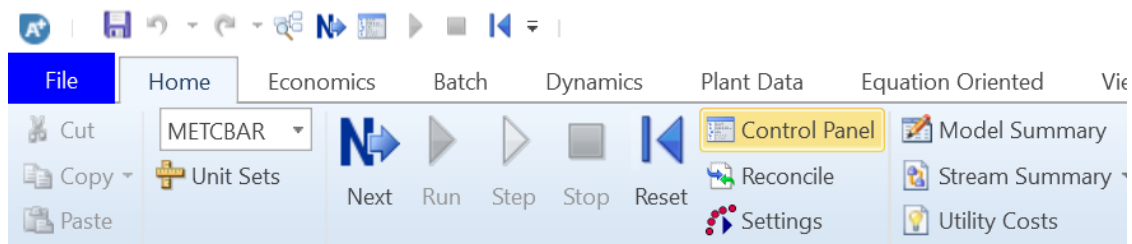
☐ Total condenser
 ☒ Partial condenser with all vapor distillate
 ☐ Partial condenser with vapor and liquid distillate

Distillate vapor fraction: **1**

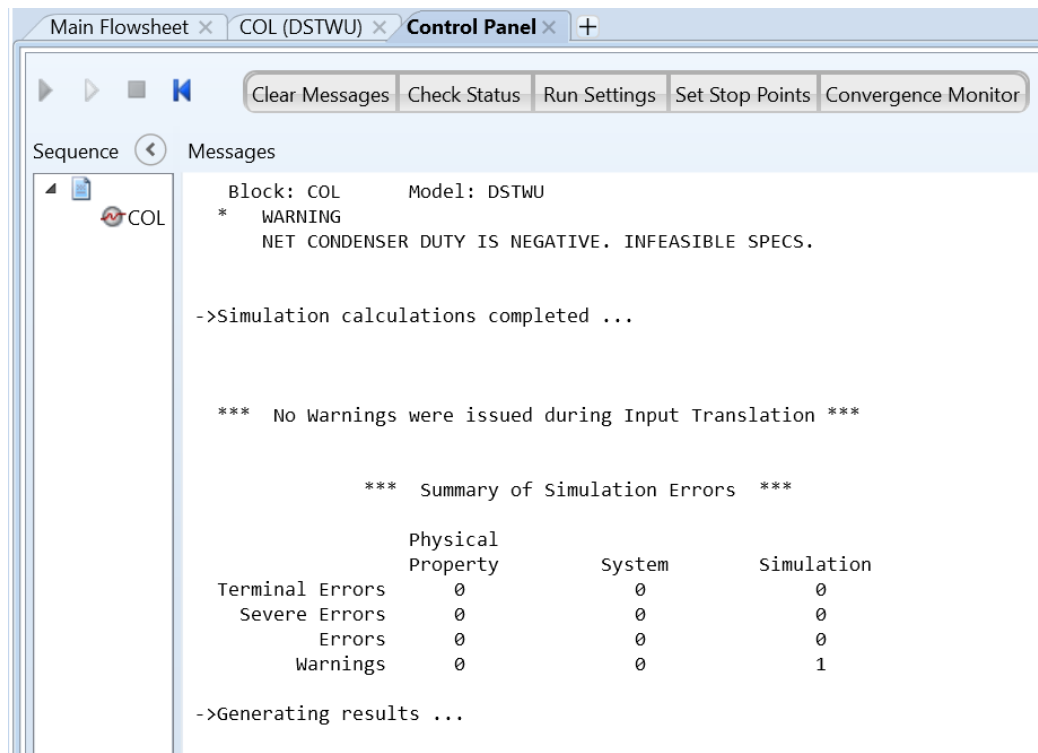
9. Open the Control Panel and run the simulation by pressing the RUN button or press F5.



10. To view the simulation status, click on Control Panel in the Home navigation pane.



There is one warning and no error messages during simulation. At the moment ignore the warning.



11. Go to the Blocks | COL | Results | Summary sheet to check the column results.

Summary Balance Reflux Ratio Profile Status		
▶ Minimum reflux ratio	3.62187	
▶ Actual reflux ratio	4.16896	
▶ Minimum number of stages	30.7436	
▶ Number of actual stages	61.4872	
▶ Feed stage	27.8237	
▶ Number of actual stages above feed	26.8237	
▶ Reboiler heating required	1.63385e+07	cal/sec
▶ Condenser cooling required	1.83027e+07	cal/sec
▶ Distillate temperature	-27.0457	C
▶ Bottom temperature	-5.48226	C
▶ Distillate to feed fraction	0.70208	
▶ HETP		

12. To check stream results, go to the Blocks | COL | Stream Results | Material sheet.

Main Flowsheet COL (DSTWU) Control Panel FEED (MATERIAL) - Results (Default) COL (DSTWU) - Stream Results			
Material	Heat	Load	Vol.% Curves Wt. % Curves Petroleum Polymers Solids
			FEED BOTT DISTL
▶ Description			
▶ From			COL COL
▶ To			COL
▶ Stream Class			CONVEN CONVEN CONVEN
▶ Maximum Relative Error			
▶ Cost Flow		\$/hr	
- MIXED Substream			
▶ Phase			Vapor Phase Liquid Phase Vapor Phase

13. To create profile of Reflux Ratio versus Number of Stages. Go Blocks | COL | Input | Calculation Options tab. Select the Generate of reflux ratio vs number of theoretical stages. Enter reasonable estimates for the initial and final number of stages, and an increment size that will give you good resolution in the resulting curve.

Options

☒ Generate table of reflux ratio vs number of theoretical stages

☐ Calculate HETP

Table of actual reflux ratio vs number of theoretical stages

Initial number of stages: 10

Final number of stages: 75

☒ Increment size for number of stages: 2

☐ Number of values in table: 11

Significant digits displayed in table: 5

HETP calculation

Packed height: meter

14. Run the simulation. To view the results table for reflux ratio versus number of stages go to Results | Reflux Ratio Profile tab.

Theoretical stages	Reflux ratio
32	34.7075
34	17.745
36	13.1437
38	9.58094
40	7.52957
42	6.71938
44	6.23334
46	5.87541
48	5.58097
50	5.31852

15. To plot the graph for Theoretical stages vs Reflux ratio, click Custom plot in the Home navigation pane.

Example 3.1.apw - Aspen Plus V12.1 - aspenONE

File Home Economics Batch Dynamics Plant Data Equation Oriented View Customize Resources

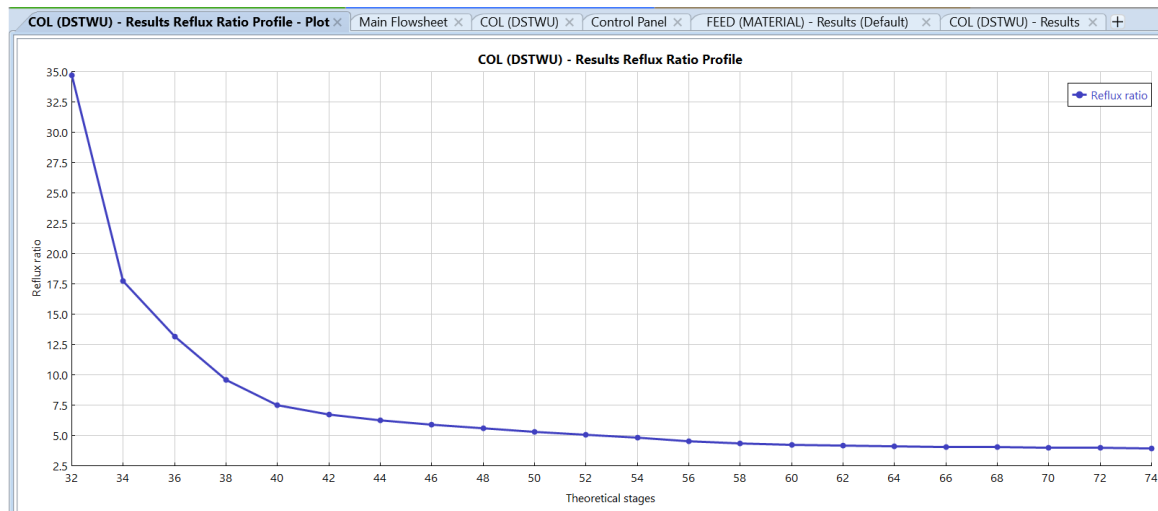
Simulation Capital: USD Utilities: USD/Year Energy Savings: MW (%) Exchangers - Unknown: 0 OK

Main Flowsheet COL (DSTWU) Control Panel FEED (MATERIAL) COL (DSTWU) - Results

Summary Balance Reflux Ratio Profile Status

Theoretical Reflux ratio

In the pop menu select Reflux ratio and click OK. The graph should appear in a new tab.



16. Increasing the number of stages increases the capital cost of the column. Whereas, increasing the reflux ratio increases the operating costs because of more material is recycled, and it must be reheated in the reboiler and recondensed in the condenser. Consequently, there is a tradeoff: larger columns have a larger upfront cost but lower operating costs, while smaller columns have smaller capital costs but are more expensive to operate. In essence, the hand-waiving argument is that a real column will need slightly more stages to achieve the separation modeled in the DSTWU block because a real column cannot achieve the same efficiency as the model.